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All groups G are finite and all reps V of G are finite-dim.

Last time. Let $\rho: G \to GL(V)$ be a rep. $\dim V = n$.
$g \mapsto (\rho(g): V \to V)$

Define $\chi_\rho: G \to \mathbb{C}$, $g \mapsto \mathrm{tr}(\rho(g))$ the character of ρ.

We have 1) $\chi_\rho(1) = n$

2) $\chi_\rho(hgh^{-1}) = \chi_\rho(g)$

3) $\chi_{\rho^*}(g) = \chi_\rho(g^{-1}) = \overline{\chi_\rho(g)}$

(G finite $\Rightarrow$ all eigenvalues $\lambda_1, \cdots, \lambda_n$ of $\rho(g)$ are roots of unity

$\Rightarrow$ eigenvalues of $\rho^*(g)$ or $\rho(g^{-1})$ are $\lambda_1^{-1}, \cdots, \lambda_n^{-1}$ ie. $\overline{\lambda_1}, \cdots, \overline{\lambda_n}$.

$\Rightarrow \chi_{\rho^*}(g) = \chi_\rho(g^{-1}) = \sum_i \overline{\lambda_i} = \overline{\chi_\rho(g)}$)

DEF A function $f: G \to G$ is called a class function if $\forall g, h \in G$, $f(hgh^{-1}) = f(g)$.

PROP Let $(\rho_1, V_1), (\rho_2, V_2)$ be the rep of G, Then

1) $\chi_{\rho_1 \oplus \rho_2}(g) = \chi_{\rho_1}(g) + \chi_{\rho_2}(g)$

2) $\chi_{\rho_1 \otimes \rho_2}(g) = \chi_{\rho_1}(g) \cdot \chi_{\rho_2}(g)$.

Pf. $(\rho_1 \oplus \rho_2)(g) = \begin{pmatrix} \rho_1(g) & \\ & \rho_2(g) \end{pmatrix} \Rightarrow 1)$

for 2). if $\rho_1(g) = (a_{ij})$ $\rho_2(g) = (b_{ij})$ then $\chi_{\rho_1}(g) = \sum_i a_{ii}$ $\chi_{\rho_2}(g) = \sum_j b_{jj}$.

$\chi_{\rho_1 \otimes \rho_2}(g) = \sum_{i,j} a_{ii} b_{jj} = \chi_{\rho_1}(g) \chi_{\rho_2}(g)$ ← diagonal entries for $v_i \otimes w_j$.

DEF For two functions $\chi, \eta: G \to \mathbb{C}$, we define

$\langle \chi, \eta \rangle := \frac{1}{|G|} \sum_{g \in G} \chi(g) \overline{\eta(g)}$ Hermitian inner product

in particular, $\langle \chi_{\rho_1}, \chi_{\rho_2} \rangle = \frac{1}{|G|} \sum_{g \in G} \chi_{\rho_1}(g) \chi_{\rho_2}(g^{-1}) \overset{h=g^{-1}}{=\!=} \frac{1}{|G|} \sum_{h \in G} \chi_{\rho_1}(h^{-1}) \chi_{\rho_2}(h) = \langle \chi_{\rho_2}, \chi_{\rho_1} \rangle$

THM (Schur's orthogonality) Let $(\rho_1, V_1), (\rho_2, V_2)$ be irreducible reps of G. Then

$$\langle \chi_{\rho_1}, \chi_{\rho_2} \rangle = \begin{cases} 1 & \text{if } (\rho_1, V_1) \simeq (\rho_2, V_2) \\ 0 & \text{if } (\rho_1, V_1) \not\simeq (\rho_2, V_2) \end{cases}$$

Recall Let $\Phi: V_1 \to V_2$ be $\mathbb{C}$-linear

Then $\widetilde{\Phi} = \frac{1}{|G|}\sum_{g\in G} \rho_2(g)\circ\Phi\circ\rho_1(g^{-1}) : V_1 \to V_2$ is G-linear.

V_1, V_2 irreducible

Schur's lemma $\Rightarrow \widetilde{\Phi} = \begin{cases} 0 & \text{if } (\rho_1,V_1)\not\simeq(\rho_2,V_2) \\ \lambda\cdot id_V & \text{if } (\rho_1,V_1)\simeq(\rho_2,V_2) \end{cases}$

$\lambda = \frac{1}{\dim V_1}\operatorname{tr}(\widetilde{\Phi}) = \frac{1}{\dim V_1}\operatorname{tr}(\Phi)$.

Pf of THM. 1) if $(\rho_1,V_1)\not\simeq(\rho_2,V_2)$, identify $V_1\simeq\mathbb{C}^m$, $V_2\simeq\mathbb{C}^n$

Consider $\Phi = E_{ij} : \mathbb{C}^m \to \mathbb{C}^n$

Then Schur's lemma $\Rightarrow \sum_{g\in G}\rho_2(g)E_{ij}\rho_1(g^{-1}) = 0$.

$\Rightarrow \forall k,l \ \sum_{g\in G}\rho_2(g)_{ks}\underbrace{(E_{ij})_{st}}_{\text{0 unless } s=i,\,t=j}\rho_1(g^{-1})_{tl} = 0 \Rightarrow \sum_{g\in G}\rho_2(g)_{ki}\rho(g^{-1})_{jl} = 0, \ \forall i,j,k,l$.

$\Rightarrow \langle\chi_{\rho_2},\chi_{\rho_1}\rangle = \frac{1}{|G|}\sum_{g\in G}\operatorname{tr}(\rho_2(g))\operatorname{tr}(\rho_1(g^{-1})) = \frac{1}{|G|}\sum_{g\in G}\sum_{i,j}\rho_2(g)_{ii}\rho_1(g^{-1})_{jj} = 0$.

2). if $(\rho_1,V_1)\simeq(\rho_2,V_2)$ identify $(\rho_1,V_1)\simeq(\rho_2,V_2)\simeq(\rho,\mathbb{C}^n)$.

Schur's lemma $\Rightarrow \frac{1}{|G|}\sum_{g\in G}\rho(g)E_{ij}\rho(g^{-1}) = \lambda\cdot id \quad \lambda = \frac{1}{\dim\mathbb{C}^n}\operatorname{tr}(E_{ij}) = \begin{cases}\frac{1}{n}, & \text{if } i=j \\ 0, & \text{if } i\neq j\end{cases}$.

$\Rightarrow \frac{1}{|G|}\sum_{g\in G}\rho(g)_{ii}\rho(g^{-1})_{jj} = \begin{cases}\frac{1}{n}, & \text{if } i=j \\ 0, & \text{if } i\neq j\end{cases}$.

$\Rightarrow \langle\chi_\rho,\chi_\rho\rangle = \frac{1}{|G|}\sum_{g\in G}\operatorname{tr}(\rho(g))\operatorname{tr}(\rho(g^{-1})) = \frac{1}{|G|}\sum_{g\in G}\sum_{i,j}\rho(g)_{ii}\rho(g^{-1})_{jj} = \sum_{i=j}\frac{1}{n} = 1$. □

REM The character of an irreducible rep is called an <u>irreducible character</u>.

Schur's orthogonality $\Rightarrow$ irreducible characters are orthonormal.

Will prove that they form an orthonormal basis of the space of class functions.

Recall any rep (ρ,V) is the direct sum of irreducibles

$V = W_1^{\oplus m_1}\oplus\cdots\oplus W_r^{\oplus m_r}$ with (ρ_i,W_i) irreducible and pairwise nonisom.

Then $\langle\chi_\rho,\chi_{\rho_i}\rangle = \langle\sum_j m_j\chi_{\rho_j},\chi_{\rho_i}\rangle = m_i$.

COR Two reps with the same characters are isomorphic.

COR For a rep (ρ,V) if $V = W_1^{\oplus m_1}\oplus\cdots\oplus W_r^{\oplus m_r}$ with (ρ_i,W_i) ——

Then $\langle\chi_\rho,\chi_\rho\rangle = m_1^2+\cdots+m_r^2 > 0$.

$\langle\chi_\rho,\chi_\rho\rangle = 1 \iff (\rho,V)$ is irreducible.

Regular representation

Recall $G \xrightarrow{\rho_{reg}} \mathbb{C}[G]$.

$$\rho_{reg}(g)\left(\sum_{x\in G} a_x x\right)=\sum_{x\in G} a_x (gx) \quad \text{Left translation}$$

$$\chi_{reg}(g) := \chi_{\rho_{reg}}(g)= tr(\rho_{reg}(g))=\begin{cases}|G|, & \text{if } g=1\\ 0, & \text{if } g\neq 1\end{cases}$$

$\Rightarrow \forall$ irreducible rep (ρ_i, W_i),

$$\langle \chi_{\rho_i}, \chi_{reg}\rangle = \frac{1}{|G|}\sum_{g\in G}\chi_{\rho_i}(g)\underbrace{\chi_{reg}(g^{-1})}_{=0 \text{ unless } g=1} = \frac{1}{|G|}\chi_{\rho_i}(1)\cdot|G| = \chi_{\rho_i}(1)=\dim W_i$$

$$\Rightarrow \mathbb{C}[G] \simeq \bigoplus_{W_i\in Irr(G)} W_i^{\oplus \dim W_i}$$

$Irr(G)=\{\text{irreducible reps of } G\}$.

$|G|=\sum_{W_i\in Irr(G)}(\dim W_i)^2$ $\qquad \sum_{x\in G} a_x x \in \mathbb{C}[G]$

Main THM Let $\mathbb{C}[G]^{Ad_G} := \{\text{class functions } f: G\to\mathbb{C}\}$

$=\{f: G\to\mathbb{C} \mid f(hgh^{-1})=f(g)\}$

$=\{f: \{\text{conjugacy classes of } G\}\to\mathbb{C}\}$ 研究表示→群的信息

Then $\{\chi_{\rho_i} \mid (\rho_i, W_i)\in Irr(G)\}$ form an orthonormal basis of $\mathbb{C}[G]^{Ad_G}$.

in particular, $\#Irr(G)=\#\{\text{conjugacy classes of } G\}$.

Pf. suffices to show that $\{\chi_{\rho_i} \mid (\rho_i, W_i)\in Irr(G)\}$ span $\mathbb{C}[G]^{Ad_G}$.

Suppose that $f\in\mathbb{C}[G]^{Ad_G}$ is orthogonal to all of χ_{ρ_i}.

✡ KEY Construction.

Given any rep (ρ, V) and any $f: G\to\mathbb{C}$ $(f\in\mathbb{C}[G])$

We define $\rho_f := \sum_{g\in G} f(g)\rho(g): V\to V$ ($\mathbb{C}$-linear) "weighted sum".

if in addition $f\in\mathbb{C}[G]^{Ad_G}$ is a class function.

$$\forall h\in G,\ \rho(h)\circ\rho_f\circ\rho(h^{-1})=\sum_{g\in G} f(g)\rho(hgh^{-1}) \xrightarrow{g'=hgh^{-1}} \sum_{g\in G} \underbrace{f(h^{-1}g'h)}_{=f(g') \text{ ← class func}}\rho(g')=\rho_f$$

$\Rightarrow \rho(h)\circ\rho_f=\rho_f\circ\rho(h), \forall h\in G \Rightarrow \rho_f: V\to V$ is G-linear.

if furthermore, V is irreducible, then $\rho_f=\lambda_f\, id_V$.

where $\lambda_f = \frac{1}{\dim V}\mathrm{tr}(\rho_f) = \frac{1}{\dim V}\sum_{g\in G} f(g)\underbrace{\mathrm{tr}(\rho(g))}_{\overline{\chi_{\rho^*}(g)}} = \frac{|G|}{\dim V}\langle f, \chi_{\rho^*}\rangle$.

Back to the proof.

$f: G\to\mathbb{C}$ is a class function. by assumption, $\forall (\rho_i, W_i)\in \mathrm{Irr}(G)$.

$\rho_f: W_i \to W_i$ is the scalar multiplication by $\frac{|G|}{\dim W_i}\langle f, \chi_{\rho_i^*}\rangle = 0$ (ρ_i^* irred)

For the regular representation $G \xrightarrow{\rho_{reg}} \mathbb{C}[G]$

$$\mathbb{C}[G] \simeq \bigoplus_{W_i\in \mathrm{Irr}(G)} W_i^{\oplus \dim W_i}$$

$\Rightarrow \rho_f: \mathbb{C}[G]\to\mathbb{C}[G]$ is zero

$\Rightarrow 0 = \rho_f(1) = \sum_{g\in G} f(g)\rho_{reg}(g)(1) = \sum_{g\in G} f(g)g \in \mathbb{C}[G] \Rightarrow f(g)=0, \forall g\in G$. □

THM. For any irreducible rep (ρ, V) of G, we have $\dim V \mid |G|$.

Pf. For each conjugacy class $\mathrm{Ad}_G(g_i)$ with g_i a representative.

Consider class function $f_i: G\to\mathbb{C}$ $\quad f_i(g) = \begin{cases} 1, & g\in \mathrm{Ad}_G(g_i) \\ 0, & g\notin \mathrm{Ad}_G(g_i)\end{cases}$

$\Rightarrow \rho_{f_i} = \sum_{g\in \mathrm{Ad}_G(g_i)} \rho(g): V\to V$ (G-linear)

V irreducible $\Rightarrow \rho_{f_i} = \lambda_i\, \mathrm{id}_V$

where $\lambda_i = \frac{1}{\dim V}\mathrm{tr}(\rho_{f_i}) = \frac{1}{\dim V}\sum_{g\in \mathrm{Ad}_G(g_i)}\mathrm{tr}(\rho(g)) = \frac{\#(\mathrm{Ad}_G(g_i))}{\dim V}\cdot \chi_\rho(g_i)$.

Claim: λ_i is an algebraic integer (root of a monic polynomial with integer coefficients)

Also $\chi_\rho(g_i^{-1})$ is an algebraic integer.
(sum of roots of unity)

$\Rightarrow \lambda_i\chi_\rho(g_i^{-1}) = \frac{\#(\mathrm{Ad}_G(g_i))}{\dim V}\chi_\rho(g_i)\chi_\rho(g_i^{-1})$ is an algebraic integer.

Summing up, $\sum_i \lambda_i\chi_\rho(g_i^{-1}) = \frac{1}{\dim V}\sum_{g\in G}\chi_\rho(g)\chi_\rho(g^{-1}) = \frac{|G|}{\dim V}\langle \chi_\rho, \chi_\rho\rangle = \frac{|G|}{\dim V}$ since (ρ, V) irreducible.

($\sum_i \lambda_i\chi_\rho(g_i^{-1})$: algebraic integer; $\frac{|G|}{\dim V}$: rational)

$\Rightarrow \frac{|G|}{\dim V}$ is an integer $\Rightarrow \dim V \mid |G|$.

Pf of Claim. Consider $\mathbb{Z}[G]$ = free abelian group generated by $G = \{\sum_{x\in G} a_x x \mid a_x\in\mathbb{Z}\}$.

$G \xrightarrow{\text{left translation}} \mathbb{Z}[G]$ $\quad l_g(\sum_{x\in G} a_x x) = \sum_{x\in G} a_x(gx)$.

Define $L_i := \sum_{g\in \mathrm{Ad}_G(g_i)} l_g: \mathbb{Z}[G]\to\mathbb{Z}[G]$ ← viewed as a matrix with integer coefficients.

P_i := characteristic polynomial of L_i (monic polynomial with integer coeff.)

Cayley-Hamilton $\Rightarrow P_i(L_i): \mathbb{Z}[G] \to \mathbb{Z}[G]$ is zero.

$P_i(L_i) = P_i\left(\sum_{g \in Ad_G(g_i)} l_g\right)$ $\qquad$ $P_i\left(\sum_{g \in Ad_G(g_i)} \rho(g)\right)$

$\Rightarrow \rho_{f_i} = \sum_{g \in Ad_G(g_i)} \rho(g): V \to V$ satisfies $P_i(\rho_{f_i}): V \to V$ is zero

$\rho_{f_i} = \lambda_i \cdot id$ $\qquad$ $P_i(\lambda_i \cdot id_V) = 0 \Rightarrow \lambda_i$ is a root of P_i. $\square$